PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Prevalence of childhood exposure to intimate partner violence in low-income and lower-middle-income countries: a systematic review
AUTHORS	Kieselbach, Berit; Kimber, Melissa; MacMillan, Harriet; Perneger, Thomas

VERSION 1 – REVIEW

REVIEWER	Narasimhan, Subasri Emory University, Hubert Department of Global Health
REVIEW RETURNED	28-May-2021

GENERAL COMMENTS	It was a pleasure to review this manuscript. I appreciate the importance to the topic and the clarity with which this systematic review was written given the disparate nature of studies in this area. Further, the supplementary material provided enough detail to follow all decisions of the review. I have no substantial edits at this time.

REVIEWER	Metheny, Nicholas University of Miami
REVIEW RETURNED	02-Jun-2021

GENERAL COMMENTS	• There Introduction overall seems a bit short and fails to cover the importance of this issue. Specifically: o there is a lack of theoretical grounding as to why witnessing IPV as a child leads to poor health outcomes o There is no discussion of how witnessing IPV as a child can lead to greater propensity for experiencing and perpetrating IPV as an adult. This is an important link in the cycle of violence that a lifecourse approach to IPV should take into account • A broader discussion of these topics, as well as potential mechanisms discussed in the literature between witnessing violence and poor health outcomes would be helpful Methods • The search strategy seems largely appropriate and the authors have substantial experience in the field of IPV to understand the literature base. However, I wonder why the authors chose not to include grey literature or data in this analysis. For example, the Demographic and Health Surveys ask questions on witnessing IPV as a child, providing an additional source of data for this study. While some studies included in this manuscript use DHS data, the authors could have supplemented the meta-analysis with data from low- and lower-middle income countries that were not represented in the literature. Similarly, government reports or other grey literature may provide additional prevalence estimates.

	• More details on the meta-analysis methods are warranted, however. What kind of random effects models were run? Were they two- or three-level models? It seems a three-level model may be necessary as subjects are nested within studies and studies within countries. Perhaps the authors can expand this section. Results • The results seem appropriate for the data analyzed Discussion • More in-depth discussion of the potential drivers of these estimates, the gaps in knowledge, and lessons learned from the included papers would be helpful.

REVIEWER	Liel, Christoph German Youth Institute, Family and Family Policies
REVIEW RETURNED	18-Jun-2021

GENERAL COMMENTS	The manuscript focuses the estimate of prevalence rates of childhood exposure to IPV in low income and lower middle-income countries in the world by using methods of meta-analysis. This is an important topic to connecting IPV as an adverse childhood experience closer with child maltreatment epidemiology and to describing a baseline for these countries as an increase of IPV during the covid-19 pandemic is discussed globally. Due to its high quality in analysing and describing the data, I have some recommendations, mostly minor in nature, targeting the way of presentation the content. Overall, as the proportion of children exposed to IPV is the study objective, I was wondering what is meant by lifetime prevalence rates? Arent they depending in child ages? The authors highlight a lack of consensus around measurements as a limitation of findings. Therefore, I would like to be informed about the ways of measurements have been included in the metha-analysis (for an overview in child maltreament data measurments see e.g., Fluke et al., 2020). This is important as the lifetime prevalence rates might highly differ depending on retrospective studies in adults or child/parental self-report studies for different childhood ages. Therefore, the inclusing criteria of studies should be described here. Is there a need to differentiate the data-analisis by measurement? Please define the lifetime prevalence in context of childhood. Detailed recommendations by sections: Abstract: - Please point out, which prevalence rates were measured, e.g. “past-year prevalence rate was 35% [in children] (…). The liefetime prevalence [of child exposure to IPV in adults?] was …” (p.3, l.31). - Keywords: Shouldn't "children" or "childhood" be added to any keyword? Introduction: - As the authors distinguish between oberservation and mere awareness of IPV, I was asking myself if direct involvement of the child should not be included to direct obervation as well (p.3, l.20). There are many constellations of IPV, in which the child is direct involved by trying to settle the conflict or the father instrumentalizes him or her to hurt the mother.

	- Due to the later operationalisation in the method section (p.4, l.4), it is not clear, what “mere awareness” decribes in detail (p.3, l.20). You may want to bypass the discourse to what extent children do be able to indirectly witnessing violence. Maybe better refer to the studies in abused mothers, that most children do witness the violence between parents at least indicretly? - The explanation of the negative impact of IPV in children reads a little static. However, the the growing-up of children is a dynamic process. Please point out, that the exposure to IPV is a highly risk (depending on age, dose and duration of violence) to negative impacting child development in a broad range of areas (emotional, affetive, cognitive, educational). - “In recent years, however, primary prevalence studies have become available and data have been collected in the context of general health surveys or surveys directly focused on violence and abuse [in women?]” (p.3, l.34). Please add references as well. Methods - Again: Please concretize the target population “IPV among children and adults” (p.3, l.52) - As the outcome of interest was defined by the keywords “exposure” or “witnessing” of IPV, maybe there is no need to distinguish here between direct and indirect involvement. - For the study, it might be helpful to be informed about the “World Bank country and lending classification” (p.4, l.42) - “No studies were excluded based on [the high?] risk of bias rating” (p.5 l.22). Results of bias assessment remained unclear at this point (desribed later in the results section). Results - Due to the total of 63 studies: “[Most All?] studies reported applying some form of random sampling (n= 63)” (p.6 l52). - As I’m familiar with meta-analysis, the number of studie is named with “k” instead of “n” to destingush to the included poupulation sample sizes. - The paragraph “Pooled prevalence estimates…” (p.7l.12) is to be inclued in the methods section. - If there is a need for shortening due to the manuscript length: “Only two studies with few participants collected data on past-month prevalence, so we did not pool these estimates” (p.9 l.12). Otherwise, please include past month prevalence as an additional research topic. Discussion - In total 67 studies is conflicting with the results section (p.7 l.54). - Overall suggestion for a deeper consideration of findings: As you measured prevalence rates in low and low middle countries the discussion section might provide by wether your findings support the connection of IPV with poverty or not. - Secondly, the education in women and girls is one major goal of prevention strategies against IPV (e.g., by the WHO), escpecially in low income countries. Therefore, I would suggest to consider the role of education in any way discussing your findings. - Describing factors that explain the variance between WHO regions, please distingusih betweenn content-related and methodological factors as this paragraphy reads too complex (p. 8, l.25).

	- The authors argue, that gender differences might be expectable as girls spend more time at home than boys, escpecially when traditional gender norms exist (p.8, l.42). I was wondering if those gender gender norms appears for mothers and fathers as well? Therfore, you would not expect gender differences in prevalence rates as men and boys spend are more likely to spend time out of home and the risk of IPV is increasing by daytimes when all family members are at home, esp. both parents. This is just a thought. - Research implication of findings (p.9): In self-report measurement, wouldn't more detailed information about the omitter of violence lead to reduced prevalence rates? One resasonable effort of existing measures (e.g., the JVC) is not the get too detailed at this point. This section might provide from detailed information about included measurements as suggested before. - Conslusion: Just "beeing alerted" in the light of adverse consequences for children exposed to IPV? We know about that. Therfore, I would suggest to highlight also the role of the medical professionals as part of development aid in low-income countries. Minorities: - Child Abuse & Neglect ("Journal of" to be deleted, p.4 l.31) - Please check the journal specifications for the presentation of statistical key figures (spaces before and after "=" or "p"? Number of decimal places?) - Heterogeneity (p.8, l.47) - "at home" instead of "in the home" (p. 10, l.43)

REVIEWER	Alebel, Animut Debre Markos University College of Health Science, Nursing
REVIEW RETURNED	13-Aug-2021

GENERAL COMMENTS	Title "Prevalence of childhood exposure to intimate partner violence in low-income and lower-middle-income countries: a systematic review and meta-analysis". Thank you very much for the opportunity. The title is fascinating, but it needs further work to make it more comprehensive. General comment: • This review is very superficial. It is not well explored, especially in terms of research and presentation of results. Specific comments: Methods • As per editor's suggestion, I have critically reviewed the method section and the following comments are forwarded. 1. In terms of research, I highly recommend including the EMBASE database, as it is the most abundant. Therefore, articles could be omitted because they did not search in this database. in general, I am not happy with the searching strategies, it needs father work. It would be better consult libertarian again. 2. Who did quality assessment of included studies? If you included all studies irrespective of their quality scores, what is the relevance of assessing quality? Could you explain it? 3. Please define the outcome of interest (IPV) in the method section. 4. As you stated, the included studied showed high heterogeneity (I2 = 99.67%). So, could you consider meta-regression using different factors (i.e, sample size, quality score, geography, and so on.) to identify the possible sources of heterogeneity? Unless, it could be better to report in a systematic review if I2 is >75%.

	5. The other concern in the statistical analysis is that the possible influences of studies with sample sizes (publication bias) were not investigated.

VERSION 1 – AUTHOR RESPONSE

Reviewer #1	
It was a pleasure to review this manuscript. I appreciate the importance to the topic and the clarity with which this systematic review was written given the disparate nature of studies in this area. Further, the supplementary material provided enough detail to follow all decisions of the review. I have no substantial edits at this time.	Thank you very much for the favourable review and for acknowledging the complex nature of the data in this research field.

Reviewer #2	
Introduction The Introduction overall seems a bit short and fails to cover the importance of this issue. Specifically: there is a lack of theoretical grounding as to why witnessing IPV as a child leads to poor health outcomes.	We added further research to highlight the pathway between witnessing IPV as a child and physical health outcomes, including non-communicable diseases mediated through a higher likelihood to engage in health risk behaviours; we also added information that speaks to the impacts of exposure on brain development and those impacts as a potentially mediating pathway to health and social outcomes later in life. (Page 4)
There is no discussion of how witnessing IPV as a child can lead to greater propensity for experiencing and perpetrating IPV as an adult. This is an important link in the cycle of violence that a lifecourse approach to IPV should take into account.	We have revised the Introduction to include information based on systematic reviews examining the relationship between childhood exposure to IPV and perpetration and victimization of intimate partner violence as an adult (Page 4).
A broader discussion of these topics, as well as potential mechanisms discussed in the literature between witnessing	We have added text about pathways between witnessing IPV and poor health outcomes, and explained that these are sometimes direct, and sometimes moderated e.g. by socio-economic status (Page 4).

violence and poor health outcomes would be helpful	We did not include information about theoretical concepts, such as social learning theory and attachment theories, given the need to adhere the word limit. Also, since the focus of the systematic review was prevalence, we thought that including information about mechanisms would go beyond the overall scope of the review.
Methods: The search strategy seems largely appropriate and the authors have substantial experience in the field of IPV to understand the literature base. However, I wonder why the authors chose not to include grey literature or data in this analysis. For example, the Demographic and Health Surveys ask questions on witnessing IPV as a child, providing an additional source of data for this study. While some studies included in this manuscript use DHS data, the authors could have supplemented the meta-analysis with data from low- and lower-middle income countries that were not represented in the literature. Similarly, government reports or other grey literature may provide additional prevalence estimates.	We appreciate that the searches would have been more comprehensive, if we included unpublished survey data. To do so however, would also have included all studies that have used DHS, ACE-IQ, ICAST, JVQ, CTS, CTS-2 and VACS surveys and individual national unpublished surveys. While the DHS programme makes complete datasets available, the same information would not be available for the other survey instruments, and we were concerned that this would involve including studies with incomplete information, thereby introducing risk of bias. Further, in the articles we reviewed, the publication of prevalence rates for childhood exposure to IPV was rarely the main purpose of the article. The data were often published alongside a long list of demographic data related to other topics. If the searches had been conducted by survey (for example all studies that used DHS data), we would have retrieved thousands of studies without data on childhood exposure to IPV. Mathur et al. (2021) conclude that meta-analysis performed with a large number of studies (>= 40) that includes non-headline results may largely mitigate publication bias Mathur MB, VanderWeele TJ. Estimating publication bias in meta-analyses of peer-reviewed studies: A meta-meta-analysis across disciplines and journal tiers. Research synthesis methods. 2021 Mar;12(2):176-91.
More details on the meta-analysis methods are warranted, however. What kind of random effects models were run? Were they two- or three-level models? It seems a three-level model may be necessary as subjects are nested within studies and studies within countries. Perhaps the authors can expand this section.	The survey was our unit of analysis, so we did not have any information about individual results. As we did not have the full datasets with the cluster information for the included studies, we did not have an option to run multi-level models. Our unit of analysis were prevalence estimates, sometimes disaggregated by subgroup (males/females). We used the metaprop procedure in STATA to pool proportions. Metaprop provides procedures for pooling proportions in a meta-analysis of multiple studies and provides a test of whether the summary effect measure is equal to zero and a test for heterogeneity. We have chosen to run a random effects model using the method of DerSimonian and Laird, with the estimate of heterogeneity being taken from the inverse-variance fixed-effect model.
Discussion: More in-depth discussion of the potential drivers of these estimates, the	We ran a meta-regression to learn more about potential drivers of the prevalence estimates and included sample size, median age of the sample, risk of bias rating, geographical region and data

gaps in knowledge, and lessons learned from the included papers would be helpful.	collection method into the model, but none of these variables could explain the variance. We believe that the heterogeneity is largely due a lack of clear and comprehensive definitions of the outcome of interest childhood exposure to IPV, such as the taxonomy proposed by Holden et al (2003). The datasets reflect this lack of agreement. In addition, there are only a few studies that used a similar methodology across countries (such as the DHS or the VACS surveys). Most research efforts were initiated by individual researchers, who developed measurement instruments that were targeted to their national or regional populations, which made comparison of outcome measures difficult. An international systematic global research effort would be desirable. We added some of these considerations into the research implications section of the manuscript (Page 11). Holden GW. Children exposed to domestic violence and child abuse: Terminology and taxonomy. Clinical child and family psychology review. 2003; 6(3), 151-160. https://doi.org/10.1023/a:1024906315255
Reviewer #3	
Overall, as the proportion of children exposed to IPV is the study objective, I was wondering what is meant by lifetime prevalence rates? Aren't they depending in child ages?	The two most common ways to measure prevalence of childhood exposure to IPV are lifetime prevalence and past-year prevalence. However, the vast majority of included studies measured lifetime prevalence (85 estimates vs. 6 estimates of past-year prevalence). Many retrospective studies are employed in populations from 15-45 years of age (DHS) or 13-24 years of age (VACS) whereby they capture lifetime exposure up to the age of 18 in adults and lifetime exposure until the point of being interviewed in adolescents. It is true that in surveys that interview children about their lifetime exposure up to their current age, children at an older age have had a higher likelihood of having observed IPV in their lives. However, IPV is usually not a one-off event. Rather, it is a stable experience that is repeated over time. Thus, should a family be characterized by IPV, it is likely that children within that family will witness IPV many times while growing up. Therefore we assume that the likelihood that this introduces bias is very low. We have also tested post-hoc whether there was any systematic difference in data that was collected from adults retrospectively and data collected from children and did not find any statistically significant difference.
The authors highlight a lack of consensus around	We extracted the individual items that were used in each of the studies to measure childhood-exposure to IPV and the scale they

measurements as a limitation of findings. Therefore, I would like to be informed about the ways of measurements have been included in the metaanalysis (for an overview in child maltreatment data measurements see e.g., Fluke et al.,2020). This is important as the lifetime prevalence rates might highly differ depending on retrospective studies in adults or child/parental self-report studies for different childhood ages. Therefore, the inclusion criteria of studies should be described here. Is there a need to differentiate the data-analysis by measurement? Please define the lifetime prevalence in context of childhood.	came from, if this information was provided. We also extracted for each item the type of IPV children were exposed to (physical, emotional/psychological). Most of the studies used an adaptation of the Conflict Tactics Scale (CTS): Straus, M. (1979). Measuring Intrafamily Conflict and Violence: The Conflict Tactics (CT) Scales. Journal of Marriage and Family, 41(1), 75-88. doi:10.2307/351733 Or the revised version CTS-2: Straus, M. A., Hamby, S. L., Boney-McCoy, S., & Sugarman, D. B. (1996). The Revised Conflict Tactics Scales (CTS2): Development and Preliminary Psychometric Data. Journal of Family Issues, 17(3), 283–316. https://doi.org/10.1177/019251396017003001 The items differed with regard to the type of violence they measured and the type of perpetrator. Of the studies included, we found 90% measured childhood exposure to physical violence; only 7 studies included physical and emotional violence in their measurement, but did not differentiate between the two, so it would not have been possible to distinguish by type of violence. In addition, about half (k=45) of the included studies asked whether the respondent has ever seen or heard their mother being beaten by the father. The remaining studies asked more broadly whether a parent was being beaten by the other parent or whether their caregiver(s) were ever beaten by their intimate partner. Although we extracted data on all individual items from the studies included in the review, we felt it would go beyond the purpose of the review to analyze the items that were used to measure childhood exposure to IPV. There is a recent detailed systematic review, that did exactly this and reviewed measurements of childhood exposure to IPV according to a taxonomy. Latzman, N. E., Vivolo-Kantor, A. M., Clinton-Sherrod, A. M., Casanueva, C., & Carr, C. (2017). Children's exposure to intimate partner violence: A systematic review of measurement strategies. Aggression and violent behavior, 37, 220-235. We have added additional text to clarify what we mean by lifetime prevalence in the context of childhood (Page 5).
Abstract: Please point out, which prevalence rates were measured, e.g. “past-year prevalence rate was 35% [in	We have included the specification “in children” when it comes to the past-year prevalence rate. As described above, the lifetime prevalence was measured retrospectively in adults and in older children/adolescents, therefore we felt it was more accurate to

children] (…). The lifetime prevalence [of child exposure to IPV in adults?] was …" (p.3, l.31).	say "lifetime prevalence of childhood exposure", which can cover lifetime prevalence in children up to their current age, as well as retrospective data from adults for exposure that occurred prior to the age of 18 years. Please also refer to changes made in the manuscript (Page 4).
Keywords: Shouldn't "children" or "childhood" be added to any keyword?	We have replaced the search terms "intimate partner violence exposure" by "childhood exposure to intimate partner violence" and the term "witnessing intimate partner violence" by "child witness". We nevertheless kept the terms "intimate partner violence" and "domestic violence" as these are more established.
Introduction: As the authors distinguish between observation and mere awareness of IPV, I was asking myself if direct involvement of the child should not be included to direct observation as well (p.3, l.20). There are many constellations of IPV, in which the child is direct involved by trying to settle the conflict or the father instrumentalizes him or her to hurt the mother.	We are aware that the literature proposes several methods/classifications of childhood exposure to IPV. For example, the taxonomy proposed by Holden (2003),includes: prenatal exposure, direct involvement (child intervenes), direct eyewitness, and indirect exposure). In practical terms however, we did not encounter any item that attempts to measure these aspects of exposure. We tried to capture all other aspects by "awareness of IPV", but have revised the text in the Introduction substantially to clarify and clearly introduce these constellations (Page 4).
- Due to the later operationalization in the method section (p.4, l.4), it is not clear, what "mere awareness" describes in detail (p.3, l.20). You may want to bypass the discourse to what extent children do be able to indirectly witnessing violence. Maybe better refer to the studies in abused mothers, that most children do witness the violence between parents at least indirectly?	We have revised both sections and included examples from the taxonomy proposed by Holden et. al. (2003) on what should be considered when it comes to childhood exposure to IPV. These exposures include seeing some of the consequences of the assault, being told about or overhearing a conversation about the assault, facing life changes because of the assault or being directly involved (Page 4).
The explanation of the negative impact of IPV in children reads a little static. However, the growing-up of children is a dynamic process. Please point out, that the exposure to IPV is a highly risk (depending on age, dose and duration of violence) to negative impacting child development in a broad range of areas (emotional, affective, cognitive, educational).	We revised this section to broaden the consequences and to make it clear that these vary depending on the factors outlined by Reviewer #3 (Page 4).
In recent years, however, primary prevalence studies have become available and data have been collected in the context of general health surveys or surveys directly focused on violence and abuse	Thank you for this comment. We would like to clarify that the "surveys directly focused on violence and abuse" refer to the Violence against Children Surveys (VACS) and the WHO Multi-Country Study on Women's Health and Domestic Violence against Women. We have added the references for these surveys (Page 4).

[in women?]" (p.3, l.34). Please add references as well.	
Methods- Again: Please concretize the target population "IPV among children and adults" (p.3, l.52)	The term "childhood exposure to intimate partner violence" refers to 1) adults, who have been asked about exposure to IPV at any point in their lives up to the age of 18 years and 2) children who have been asked about exposure to IPV at any point in their lives up to the time of the survey. We added some explanatory text to clarify the target population (Page 5).
As the outcome of interest was defined by the keywords "exposure" or "witnessing" of IPV, maybe there is no need to distinguish here between direct and indirect involvement.	We have revised the definition, given that the construct has already been defined in the introduction (Page 5).
For the study, it might be helpful to be informed about the "World Bank country and lending classification" (p.4, l.42)	The World Bank assigns the world's economies into four income groups—low, lower-middle, upper-middle, and high-income countries. The classifications are updated each year on July 1 and are based on Gross National Income per capita in current USD of the previous year. As this classification is used frequently, and we are limited by manuscript word length, we considered it sufficient to add the reference to the World Bank classification website, instead of adding the definition into the text. (Page 5).
"No studies were excluded based on [the high?] risk of bias rating" (p.5 l.22). Results of bias assessment remained unclear at this point (described later in the results section).	The risk of bias was assessed with the. risk of bias tool for prevalence studies developed by Hoy and colleagues: Hoy D, Brooks P, Woolf A, Blyth F, March L, Bain C, et al. Assessing risk of bias in prevalence studies: modification of an existing tool and evidence of interrater agreement. Journal of clinical epidemiology 2012;65(9):934-9. https://doi.org/10.1016/j.jclinepi.2011.11.014 Of the 62 studies that were reviewed, 54 had a low risk of bias, 8 had a moderate risk of bias and no studies had a high risk of bias. As no studies were rated high risk of bias, we did not exclude any. Please also refer to the discussion about the risk of bias assessment further below. We have added some text in the Methods section (Page 5) and the Results section (Page 7) to further clarify.
Results Due to the total of 63 studies: "[Most All?] studies reported	We changed the text to "Almost all studies…" and mentioned k (Page 7).

applying some form of random sampling (n= 63)" (p.6 l52).	
As I'm familiar with meta-analysis, the number of studies is named with "k" instead of "n" to distinguish to the included population sample sizes.	We have changed the number of studies to k throughout the document and in the flowchart. We kept n, where we refer to the number of outcomes/estimates, as some studies were multi-country studies and provided multiple prevalence estimates. We use N, when we refer to the total population.
- The paragraph "Pooled prevalence estimates…" (p.7l.12) is to be included in the methods section.	Moved the paragraph to the Methods section (Page 6).
- If there is a need for shortening due to the manuscript length: "Only two studies with few participants collected data on past-month prevalence, so we did not pool these estimates" (p.9 l.12). Otherwise, please include past month prevalence as an additional research topic.	Good suggestion. We have deleted all reference to past month prevalence and adapted the total number of studies accordingly.
Discussion - In total 67 studies is conflicting with the results section (p.7 l.54).	Corrected (Page 8).
- Overall suggestion for a deeper consideration of findings: As you measured prevalence rates in low and low middle countries the discussion section might provide by whether your findings support the connection of IPV with poverty or not.	Poverty and exposure to intimate partner violence can be linked at two levels. Firstly, the socio-economic status of a country might be related to lack of funding for health, social and legal services and systems, which can impact levels of violence. Second, poverty at the family level, is often related to unemployment, lower levels of education, and settlements in areas where there is more violence and crime; these are known risk factors for childhood exposure to parental violence. While we had information about the socio-economic status of the countries, we did not have data about the economic status or poverty at the family level. We included some considerations about the linkages between poverty and violence in the Discussion (Page 9).
- Secondly, the education in women and girls is one major goal of prevention strategies against IPV (e.g., by the WHO), especially in low	We included an additional paragraph in the practice and policy implications focusing on evidence-strategies for the primary prevention of IPV, including education of women and girls and economic empowerment of women (Page 11).

income countries. Therefore, I would suggest to consider the role of education in any way discussing your findings.	
Describing factors that explain the variance between WHO regions, please distinguish between content-related and methodological factors as this paragraph reads too complex (p. 8, l.25).	This text has now been substantially revised (Page 10)
The authors argue, that gender differences might be expectable as girls spend more time at home than boys, especially when traditional gender norms exist (p.8, l.42). I was wondering if those gender norms appear for mothers and fathers as well? Therefore, you would not expect gender differences in prevalence rates as men and boys spend are more likely to spend time out of home and the risk of IPV is increasing by daytimes when all family members are at home, esp. both parents. This is just a thought.	IPV is more likely to happen if both partners are at home, as it is not often perpetrated in public. This statement is supported by research examining spatial and temporal patterns of violence, which often shows an increase during holidays for example. Independent of the question of whether or not the perpetrator is at home, we wanted to point out that in many settings, there could be systematic bias related to a higher level of prevalence among girls given locally-specific cultural norms that result in girls spending more time at home (for example, because boys are given priority to pursue education or out-of-home work) and therefore, have a greater likelihood of witnessing IPV.
Research implication of findings (p.9): In self-report measurement, wouldn't more detailed information about the emitter of violence lead to reduced prevalence rates? One reasonable effort of existing measures (e.g., the JVC) is not the get too detailed at this point. This section might provide from detailed information about included measurements as suggested before.	We deleted the suggestion to ask more detailed information about the offender (Page 11).
Conclusion: Just "being alerted" in the light of adverse consequences for children exposed to IPV? We know about that. Therefore, I would suggest to highlight also the role of the medical professionals as part of development aid in low-income countries.	We revised the text to highlight the important role of health care providers (Page 13).

Minorities: - Child Abuse & Neglect ("Journal of" to be deleted, p.4 l.31) - Please check the journal specifications for the presentation of statistical key figures (spaces before and after "=" or "p"? Number of decimal places?) - Heterogeneity (p.8, l.47) - "at home" instead of "in the home" (p. 10, l.43	We appreciate the Reviewer identifying the need for these corrections. All points have been addressed.
Reviewer #4	
In terms of research, I highly recommend including the EMBASE database, as it is the most abundant. Therefore, articles could be omitted because they did not search in this database. In general, I am not happy with the searching strategies, it needs further work. It would be better consult libertarian again.	We appreciate the Reviewer's recommendation, however we favoured searching some of the specialized databases such as CINAHL and PsycINFO, who include more papers on child abuse and neglect, compared to EMBASE, to account for the fact that the topic does not have a tradition in in the biomedical field. The search strategy has been reviewed in two iterations by an information scientist (senior library scientist). We modeled our search on strategies that have been used by other prevalence studies in the field of child maltreatment. These include for example: Moody, G., Cannings-John, R., Hood, K., Kemp, A., & Robling, M. (2018). Establishing the international prevalence of self-reported child maltreatment: a systematic review by maltreatment type and gender. BMC public health, 18(1), 1-15. Stoltenborgh, M., Van Ijzendoorn, M. H., Euser, E. M., & Bakermans-Kranenburg, M. J. (2011). A global perspective on child sexual abuse: Meta-analysis of prevalence around the world. Child maltreatment, 16(2), 79-101. Stoltenborgh, M., Bakermans-Kranenburg, M. J., & Van Ijzendoorn, M. H. (2013). The neglect of child neglect: a meta-analytic review of the prevalence of neglect. Social psychiatry and psychiatric epidemiology, 48(3), 345-355. We also empirically tested the search strategy against a set of 10 pre-defined papers. While we acknowledge that we could potentially have retrieved a few additional relevant papers, we have no reason to believe that not searching EMBASE led to systematic bias in the prevalence estimates. Comparisons of search strategies that included EMBASE with those that did not include them in intervention studies, has found only marginal impacts on the overall result of the studies:

	Sampson M. Should meta-analysts search Embase in addition to Medline? Journal of Clinical Epidemiology. 2003;56(10):943-55. Halladay CW, Trikalinos TA, Schmid IT, Schmid CH, Dahabreh IJ. Using data sources beyond PubMed has a modest impact on the results of systematic reviews of therapeutic interventions. J Clin Epidemiol. 2015;68(9):1076-84.
Who did quality assessment of included studies? If you included all studies irrespective of their quality scores, what is the relevance of assessing quality? Could you explain it?	The risk of bias was assessed using the risk of bias tool for prevalence studies by Hoy and colleagues: Hoy D, Brooks P, Woolf A, Blyth F, March L, Bain C, et al. Assessing risk of bias in prevalence studies: modification of an existing tool and evidence of interrater agreement. Journal of clinical epidemiology 2012;65(9):934-9. https://doi.org/10.1016/j.jclinepi.2011.11.014 The quality assessment of the studies was done by the first author and the second author independently. The inter-rater reliability was k=.78. Of the 62 studies that were reviewed, 54 had a low risk of bias, 8 had a moderate risk of bias and no studies were classified as high risk of bias, which is why we did not exclude any. Lastly, we also stratified prevalence estimates by the risk of bias rating and did not find any differences (Page 8 of the former manuscript, now replaced by meta-regression). We are however aware that the tool has limitations, which we discussed in some detail in the Discussion (Page 11) and that the development of a “gold standard” for quality assessments of prevalence studies is overdue.
Please define the outcome of interest (IPV) in the method section.	We added the definition of IPV in addition to the outcome of interest, childhood exposure to IPV (Page 5).
As you stated, the included studies showed high heterogeneity (I2 = 99.67%). So, could you consider meta-regression using different factors (i.e, sample size, quality score, geography, and so on.) to identify the possible sources of heterogeneity? Unless, it could be better to report in a systematic review if I2 is >75%.	We followed the reviewers suggestion and replaced the prior stratification by risk of bias rating and by sample size with a meta-regression. Sample size, median age of the sample, risk of bias rating, geographical region and data collection method (household, school) were entered into a meta-regression. None of the independent variables was associated with prevalence, all p-values were non-significant. We have amended the method section accordingly. High heterogeneity is very common in interpersonal violence research, both in prevalence studies (Moody, Cannings-John, Hood, Kemp, & Robling, 2018; Stoltenborgh, Bakermans-Kranenburg, Alink, & van Ijzendoorn, 2015; Stoltenborgh, Van Ijzendoorn, Euser, & Bakermans-Kranenburg, 2011; Yon, Mikton, Gassoumis, & Wilber, 2017) and in intervention studies (Hockenhull et al., 2015) given that violence is a complex and heterogenous phenomenon, which is also reflected in the data.

	Except for a few international efforts such as the DHS, which include a limited number of optional items on interpersonal violence, nationally or internationally funded efforts to collect data systematically are lacking. Global research effort is limited to individual researchers, who often have developed designs relevant to their national populations. We still believe it is important to synthesize prevalence data using the existing measures; given the large number of surveys and the face validity of existing measures as they correspond with data on IPV prevalence. Hockenhull, J., Cherry, M., Whittington, R., Dickson, R., Leitner, M., Barr, W.,& McGuire, J. (2015). Heterogeneity in interpersonal violence outcome research: an investigation and discussion of clinical and research implications. Aggression and violent behavior, 22, 18-25. Moody, G., Cannings-John, R., Hood, K., Kemp, A., & Robling, M. (2018). Establishing the international prevalence of self-reported child maltreatment: a systematic review by maltreatment type and gender. BMC Public Health, 18(1), 1164-1164. doi:10.1186/s12889-018-6044-y Stoltenborgh, M., Bakermans-Kranenburg, M. J., Alink, L. R. A., & van Ijzendoorn, M. H. (2015). The Prevalence of Child Maltreatment across the Globe: Review of a Series of Meta-Analyses. Child Abuse Review, 24(1), 37-50. doi:10.1002/car.2353 Stoltenborgh, M., Van Ijzendoorn, M. H., Euser, E. M., & Bakermans-Kranenburg, M. J. (2011). A global perspective on child sexual abuse: meta-analysis of prevalence around the world. Child maltreatment, 16(2), 79-101. Retrieved from https://journals.sagepub.com/doi/pdf/10.1177/1077559511403920 Yon, Y., Mikton, C. R., Gassoumis, Z. D., & Wilber, K. H. (2017). Elder abuse prevalence in community settings: a systematic review and meta-analysis. The lancet global health, 5(2), e147-e156.
The other concern in the statistical analysis is that the possible influences of studies with sample sizes (publication bias) were not investigated.	To investigate publication bias further, we used a funnel plot and Egger's regression test at a 5% significance level. Visual inspection of the funnel plot revealed that the asymmetry was caused by smaller studies, which tended to give results emphasizing higher prevalence rates. Egger's regression test was significant (p=0.027). We finally applied the trim and fill method to calculate whether potential publication bias had an impact on the pooled prevalence estimates. Seven additional studies were imputed, but they did not affect the summary estimate. The Methods and Results sections were amended accordingly (Page 6 and 9)-

VERSION 2 – REVIEW

REVIEWER	Liel, Christoph German Youth Institute, Family and Family Policies
REVIEW RETURNED	13-Oct-2021

GENERAL COMMENTS	My congratulation, the manuscript has benefited really from the revisions. All of my recommendations have been addressed in appropriate ways. The procedure of the systematic review and results are well described and discussed now. Therefore, I recommend the publication of this study, after two minorities were fixed: - “mixed[1]-sex samples” (p.5 l.6) - Delete 0 in statistic codes (twice k= on p.7 and twice p= on p.7/8) or include 0 in all codes

REVIEWER	Alebel, Animut Debre Markos University College of Health Science, Nursing
REVIEW RETURNED	25-Sep-2021

GENERAL COMMENTS	I would like to thank the author being submitting the revised version of their manuscript. The author tried to justify some of my concerns. However, I still have a few concerns that must be addressed. 1. The paper also needs a language edition. 2. The introduction still needs some work. For example, the authors should justify why they want to conduct a systematic review. It requires a clear justification and why also in low and lower-middle-income countries. I am not happy with the introduction. It needs major revision. 3. Why did you excluded unpublished data? This is one source of publication bias. 4. Still, I am worried that the authors want to estimate the pooled proportion despite very high heterogeneity (I2 =99.7%). If heterogeneity is ≥ 75%, it is better to report in a systematic review rather than meta-analysis. In addition, the authors mentioned that included primary studies used different measurements to assess childhood IPV. Thus, it is appropriate to sum up these studies together.

VERSION 2 – AUTHOR RESPONSE

Reviewer #3	
My congratulation, the manuscript has benefited really from the revisions. All of my recommendations have been addressed in appropriate ways. The procedure of the systematic review and results are well described and discussed now. Therefore, I recommend the publication of this study, after two minorities were fixed: - "mixed[1]-sex samples" (p.5 l.6) - Delete 0 in statistic codes (twice k= on p.7 and twice p= on p.7/8) or include 0 in all codes	Thank you very much for the favourable review and for the thorough review of our revised manuscript. We appreciate the identification of these two remaining issues, which have been revised.
Reviewer #4	
The paper also needs a language edition	Thank you very much for your careful additional reviews, and the helpful comments. We had the paper thoroughly edit. Please see attached a track-change version for a detailed overview of the edits.
The introduction still needs some work. For example, the authors should justify why they want to conduct a systematic review. It requires a clear justification and why also in low and lower-middle-income countries. I am not happy with the introduction. It needs major revision.	There is currently no synthesized knowledge about the prevalence of childhood exposure to intimate partner violence, in particularly not for low- and middle-income countries. For all other forms of child maltreatment, such figures exist. Child maltreatment is a neglected and very underfunded field of research. It is helpful for advocates, policy makers and planners to have an overview about prevalence of childhood exposure to intimate partner violence around the world. We focus on low- and middle-income countries, because a review on high-income countries already exists and we wanted to address the data gap with respect to low- and middle-income countries. We have amended the text for additional clarity (please see page 3).
Why did you excluded unpublished data? This is one source of publication bias.	We did not try to locate unpublished data. Our aim was to use a replicable search strategy, so decided against including unpublished datasets. We aimed to include data that has been screened/validated by the publication process. Given that this is a study about prevalence, and we are not aware of any financial, intellectual or programmatic interests that could be involved in children's exposure to intimate partner violence, we do not expect that publication bias plays a major role in this field. We have also examined the included papers in detail. Of the 100 studies we analyzed, only 4 included the primary purpose of publishing the prevalence of childhood exposure to intimate partner violence, so could have been subject to publication bias. All other studies included in our review had another primary research focus. Prevalence data on childhood exposure to IPV

	was mainly included among other demographic or epidemiological data as background information or in an appendix. To investigate publication bias further, we used a funnel plot and Egger's regression test at a 5% significance level. Visual inspection of the funnel plot revealed that the asymmetry was caused by smaller studies, which tended to give results emphasizing higher prevalence rates. Egger's regression test was significant (p=0.027). In addition, we applied the trim and fill method to calculate whether potential publication bias had an impact on the pooled prevalence estimates. Seven additional studies were imputed, but they did not affect the summary estimate.
Still, I am worried that the authors want to estimate the pooled proportion despite very high heterogeneity (I2 =99.7%). If heterogeneity is ≥ 75%, it is better to report in a systematic review rather than meta-analysis. In addition, the authors mentioned that included primary studies used different measurements to assess childhood IPV. Thus, it is appropriate to sum up these studies together.	We acknowledge, that in the presence of heterogeneity, the pooled proportion is merely the average proportion. We still believe that this is a useful descriptive statistic for the reader, but as you have pointed out, it cannot be assumed to apply everywhere. We have included additional text to ensure that the pooled average is not presented as a universally valid value, and also removed meta-analysis from the title of the paper (see amended text in the Title, Abstract, and major changes in the Discussion Pages 8-9) We nevertheless would like to point out that high heterogeneity is very common in interpersonal violence research, including meta-analyses, both in prevalence studies (Moody, Cannings-John, Hood, Kemp, & Robling, 2018; Stoltenborgh, Bakermans-Kranenburg, Alink, & van Ijzendoorn, 2015; Stoltenborgh, Van Ijzendoorn, Euser, & Bakermans-Kranenburg, 2011; Yon, Mikton, Gassoumis, & Wilber, 2017) and in intervention studies (Hockenhull et al., 2015) given that violence is a complex and heterogenous social phenomenon, and often no comparable measurement standards exist, which is also reflected in the heterogeneity of the data. Hockenhull, J., Cherry, M., Whittington, R., Dickson, R., Leitner, M., Barr, W.,& McGuire, J. (2015). Heterogeneity in interpersonal violence outcome research: an investigation and discussion of clinical and research implications. Aggression and violent behavior, 22, 18-25. Moody, G., Cannings-John, R., Hood, K., Kemp, A., & Robling, M. (2018). Establishing the international prevalence of self-reported child maltreatment: a systematic review by maltreatment type and gender. BMC Public Health, 18(1), 1164-1164. doi:10.1186/s12889-018-6044-y Stoltenborgh, M., Bakermans-Kranenburg, M. J., Alink, L. R. A., & van Ijzendoorn, M. H. (2015). The Prevalence of Child Maltreatment across the Globe: Review of a Series of Meta-Analyses. Child Abuse Review, 24(1), 37-50. doi:10.1002/car.2353 Stoltenborgh, M., Van Ijzendoorn, M. H., Euser, E. M., & Bakermans-Kranenburg, M. J. (2011). A global perspective on child sexual abuse: meta-analysis of prevalence around the

world. Child maltreatment, 16(2), 79-101. Retrieved from https://journals.sagepub.com/doi/pdf/10.1177/1077559511403920

Yon, Y., Mikton, C. R., Gassoumis, Z. D., & Wilber, K. H. (2017). Elder abuse prevalence in community settings: a systematic review and meta-analysis. The lancet global health, 5(2), e147-e156.

VERSION 3 – REVIEW

REVIEWER	Alebel, Animut Debre Markos University College of Health Science, Nursing
REVIEW RETURNED	26-Jan-2022

GENERAL COMMENTS	The paper have greatly improved and suitable for publication.